50 Top Databricks Interview Questions

Data Engineer Interview Preparation — Spark, Delta Lake, Unity Catalog, SQL & Production

Interview tip: *Be ready to explain concepts with a real project example and discuss performance, scalability, cost, security, and failure handling.*

Databricks Fundamentals

1. What is Databricks, and how is it different from a traditional data warehouse?

2. Explain the Databricks Lakehouse architecture.

3. What are the key components of Databricks?

4. What is the difference between a Databricks workspace, cluster, and SQL warehouse?

5. What are notebooks in Databricks, and how are they used in data engineering?

6. What is Databricks Runtime (DBR), and why is the runtime version important?

7. What is the difference between all-purpose clusters and job clusters?

8. What are cluster policies, and why are they useful?

Apache Spark & Performance

9. Explain Spark's architecture: driver, executors, tasks, and stages.

10. What is lazy evaluation in Spark?

11. What is the difference between transformations and actions?

12. What is the difference between narrow and wide transformations?

13. What causes a shuffle in Spark, and why can it be expensive?

14. What is partitioning in Spark?

15. What is the difference between repartition() and coalesce()?

16. How would you troubleshoot a slow Spark job?

17. What is data skew, and how can you handle it?

18. When would you use broadcast joins?

19. What is caching/persisting in Spark, and when should you use it?

Delta Lake

20. What is Delta Lake, and why is it important in Databricks?

21. How does Delta Lake provide ACID transactions?

22. What is the Delta transaction log (_delta_log)?

23. What is the difference between a managed Delta table and an external Delta table?

24. What is Delta Lake time travel? Give a practical use case.

25. What is the difference between RESTORE and time travel?

26. What is OPTIMIZE in Delta Lake?

27. What is Z-ORDER, and when should it be used?

28. What is VACUUM, and what risks should you consider before running it?

29. What is schema enforcement vs schema evolution in Delta Lake?

30. How do MERGE, INSERT, UPDATE, and DELETE work with Delta tables?

31. How would you implement an incremental load using Delta Lake?

Databricks SQL & Data Modeling

32. What is the difference between a view, temporary view, and materialized view in Databricks?

33. How do you implement SCD Type 1 in Databricks?

34. How do you implement SCD Type 2 using Delta Lake?

35. How would you remove duplicate records from a Delta table?

36. How do window functions help in data engineering use cases?

37. How would you identify and optimize an expensive SQL query in Databricks?

Unity Catalog & Governance

38. What is Unity Catalog, and why is it used?

39. Explain the Unity Catalog hierarchy: metastore, catalog, schema, and table.

40. What is the difference between managed and external tables under Unity Catalog?

41. How does Unity Catalog handle access control and permissions?

42. What is data lineage in Unity Catalog?

43. What are external locations and storage credentials?

44. How would you securely connect Databricks to cloud storage?

Jobs, Workflows & Production

45. What are Databricks Workflows/Jobs, and how do you schedule a production pipeline?

46. How do you pass parameters between Databricks jobs and notebooks?

47. How do you handle failures and retries in Databricks workflows?

48. What is job cluster reuse, and how can you reduce compute cost?

49. How would you monitor a production Databricks pipeline?

50. How do you implement CI/CD for Databricks notebooks and jobs?

51. How would you migrate a legacy ETL pipeline such as Talend/ADF to Databricks?

52. Design an end-to-end Databricks pipeline for ingesting files, transforming data, and loading curated Delta tables.